

1. Which neighbourhoods have the highest average price?

Query:

```
SELECT neighbourhood_cleansed,  
count(*) as Total_Listings,  
round(avg(CAST(REPLACE(REPLACE(price, '$', ''), ',', '')) AS REAL)),2) as  
Average_Price  
FROM listings  
GROUP BY neighbourhood_cleansed  
ORDER BY Average_Price DESC;
```

Output:

	neighbourhood_cleansed	Total_Listings	Average_Price
1	Near North Side	1028	2925.76
2	Loop	593	1501.95
3	West Town	777	289.87
4	Archer Heights	7	289.6
5	Lincoln Park	293	281.69
6	Near South Side	260	253.83
7	Near West Side	607	252.86
8	Lake View	549	246.53
9	Clearing	11	227.91
10	Avalon Park	5	203.2
11	Oakland	19	203.19
12	North Center	136	193.03
13	Beverly	19	188.44
14	Grand Boulevard	152	186.7

Analysis: Near North Side has the highest average listing price at \$2,925.76, with over 1,000 listings, making it both the most expensive and one of the most active neighbourhoods. Loop follows at \$1501.95, also with a high listing count of 593. Notably, Archer Heights and Clearing have very few listings, so their average prices may not be representative. The majority of neighbourhoods fall in the \$180-290 range, suggesting Near North Side and Loop are clear premium outliers.

Note: Unlike the room type price query, this query was not filtered for outliers, so extreme listings may be influencing the averages in some neighbourhoods.

2. Which neighbourhoods get the best reviews?

Query:

```
SELECT neighbourhood_cleansed,  
count(*) as Total_Listings,  
round(avg(review_scores_rating), 2) as Average_Review  
FROM listings  
GROUP BY neighbourhood_cleansed  
ORDER BY Average_Review DESC;
```

OutPut:

	neighbourhood_cleansed	Total_Listings	Average_Review
1	Ashburn	9	4.96
2	Forest Glen	6	4.95
3	Burnside	3	4.95
4	Hegewisch	6	4.93
5	Pullman	13	4.91
6	West Lawn	12	4.9
7	Beverly	19	4.89
8	South Deering	8	4.88
9	North Park	22	4.88
10	Mount Greenwood	2	4.88
11	Edison Park	3	4.88
12	Irving Park	189	4.87
13	Belmont Cragin	48	4.86
14	North Center	136	4.85

Analysis: Ashburn has the highest average review score at 4.96, however it only has 9 total listings, making it a less reliable sample. Most of the top-rated neighbourhoods have very few listings, which can inflate average scores. Irving Park and North Center are notable expectations, with 189 and 136 listings respectively and strong average ratings of 4.87 and 4.85, making them the most statistically reliable top performers. Overall review scores are consistently high across all neighbourhoods shown, ranging narrowly between 4.85 and 4.96.

3. What room type gets the best reviews?

Query:

```
Select room_type,  
count(*) as Total_Listings,  
round(avg(review_scores_rating), 2) as Average_Review  
FROM listings  
GROUP BY room_type  
ORDER BY Average_Review DESC;
```

Output:

	room_type	Total_Listings	Average_Review
1	Entire home/apt	6668	4.79
2	Hotel room	117	4.68
3	Private room	1839	4.67
4	Shared room	36	4.65

Analysis: Entire home/apt listings receive the highest average review score at 4.79 and are by far the most common room type, with 6,668 listings, making this the most reliable and popular category. Private rooms come in third with an average score of 4.67 across 1,839 listings, also a solid sample size. Hotel rooms and shared rooms have significantly fewer listings at 117 and 36, respectively, making their averages less conclusive. Overall, the difference in review scores across all room types is fairly narrow, suggesting guests are generally satisfied regardless of room type.

4. Which neighbourhood has the most demand (low availability)?

Query:

```
SELECT neighbourhood_cleansed,  
count(*) as Total_Listings,  
round(avg(availability_30), 2) as Average_Avaiability_in_30_Days  
FROM listings  
GROUP BY neighbourhood_cleansed  
ORDER BY Average_Avaiability_in_30_Days ASC;
```

OutPut:

	neighbourhood_cleansed	Total_Listings	Average_Avaiability_in_30_Days
1	Gage Park	6	7.67
2	Edgewater	174	9.26
3	Lincoln Square	115	10.42
4	Logan Square	463	10.51
5	Woodlawn	222	10.77
6	South Chicago	36	10.81
7	Lincoln Park	293	11.42
8	Garfield Ridge	11	11.64
9	Kenwood	81	11.7
10	West Town	777	11.79
11	Avondale	193	12.04
12	Lower West Side	235	12.17
13	North Center	136	12.57
14	Irving Park	189	12.66

Analysis: Gage Park has the lowest availability at 7.67 days out of 30, suggesting the highest demand, however with only 6 listings, this may not be representative. Edgewater is a more reliable indicator with 174 listings and an average availability of just 9.26 days. Logan Square and West Town stand out as high-demand neighbourhoods with an average of 10.5-11.8 available days. Overall, the neighbourhoods at the top of the list tend to have low availability regardless of listing count, pointing to consistently strong demand in these areas.

5. What's the price difference between entire homes and private rooms?

Query:

```
SELECT room_type,  
count(*) as Total_Listings,  
round(avg(CAST(REPLACE(REPLACE(price, '$', ''), ', ', '' ) AS REAL)), 2) as  
Average_Price  
FROM listings  
WHERE price IS NOT NULL  
AND (CAST(REPLACE(REPLACE(price, '$', ''), ', ', '' ) AS REAL)) BETWEEN 0 and  
10000  
GROUP BY room_type  
ORDER BY Average_Price DESC;
```

Output:

	room_type	Total_Listings	Average_Price
1	Hotel room	32	351.13
2	Entire home/apt	6200	246.53
3	Private room	1493	83.01
4	Shared room	33	45.18

Analysis: Entire home/apt listings average \$246.53 per night compared to \$83.01 for private rooms, meaning entire homes are roughly 3x more expensive. Hotel rooms have the highest average price at \$351.13, though the sample size is small at only 32 listings. Shared rooms are the most affordable option at \$45.18 on average. The large price gap between entire homes and private rooms suggests that guests looking for a budget-friendly option may prefer private rooms, which also have a strong listing count of 1,493. During exploration, hotel room listings were found to contain extreme outliers with some priced as high as \$50,000, which would have significantly skewed the averages. A filter was applied to exclude prices above \$10,000 to ensure more accurate and representative results.

6. Which hosts have the most listings? Are they commercial operators?

Query:

```
SELECT host_id,  
       host_name,  
       count(*) as Total_Listings  
FROM listings  
GROUP BY host_id  
ORDER BY Total_Listings DESC;
```

Output:

	host_id	host_name	Total_Listings
1	107434423	NULL	612
2	446820235	LuxurybookingsFZE	123
3	248760412	Cloud9	99
4	395498979	Milan	80
5	391945218	Matt	65
6	457438194	Anita	62
7	177188450	Level	59
8	533234561	Tina	48
9	435105119	Sonder (Chicago)	45
10	98193524	Jeremy	38
11	518917937	Bella	37
12	154245517	Assel	36
13	396270947	Jennifer	33
14	243035563	Studebecker	33

Analysis: The top host has 612 listings, but appears as NULL, suggesting missing or incomplete data for that host name. LuxurybookingsFZE and Cloud9 follow with 123 and 99 listings respectively, names that strongly suggest commercial property management companies rather than individual hosts. Milan, Matt and Anita appear to be individual hosts, but with 80, 65 and 62 listings respectively, which is still unusually high for a single person and may indicate small-scale commercial operators. Overall, the hosts with the most listings are likely commercial operators, as the volume of listings is too high to be typical of casual individual hosting.

7. What factors predict a higher review score?

Query:

```
SELECT room_type,  
round(avg(CAST(REPLACE(REPLACE(price, '$', ''), ',', '')) AS REAL)), 2) as  
Average_Price,  
round(AVG(review_scores_rating), 2) as Average_Rating,  
round(AVG(review_scores_cleanliness), 2) as Average_Cleanliness,  
round(AVG(review_scores_location), 2) as Average_Location,  
round(AVG(review_scores_value), 2) as Average_Value_Score,  
COUNT(*) as Total_Listings  
FROM listings  
WHERE Review_Scores_Rating IS NOT NULL  
AND CAST(REPLACE(REPLACE(price, '$', ''), ',', '')) AS REAL BETWEEN 10 AND 1000  
GROUP BY room_type  
ORDER BY Average_Rating DESC;
```

OutPut:

	room_type	Average_Price	Average_Rating	Average_Cleanliness	Average_Location	Average_Value_Score	Total_Listings
1	Entire home/apt	207.95	4.79	4.8	4.78	4.69	4986
2	Hotel room	340.14	4.71	4.77	4.89	4.55	29
3	Private room	78.33	4.69	4.65	4.64	4.63	1206
4	Shared room	44.72	4.65	4.62	4.68	4.65	25

Analysis: Entire home/apt listings score the highest across almost every category, rating (4.79), cleanliness (4.8), and location (4.89), while also being the most represented with 4,986 listings, making these results the most reliable. Hotel rooms score the highest for location (4.89), which makes sense given hotels are typically situated in central or convenient areas, however their value score is notably lower at 4.55, suggesting guests feel they are paying more than the experience is worth. Private rooms and shared rooms score closely across all categories, despite private rooms averaging \$78.33 versus shared rooms at \$44.72. Overall, cleanliness and location appear to be consistently strong across all room types, suggesting these factors alone may not be the key differentiators of higher review scores.

8. What percentage of listings are available year-round?

Query:

```
SELECT round(100.0 * SUM(CASE WHEN availability_365 = 365 THEN 1 ELSE 0 END)
/ count(*), 2) as Fully_Available_Percent,
round(100.0 * SUM(CASE WHEN availability_365 = 0 THEN 1 ELSE 0 END) / count(*),
2) as Fully_Booked_Percent,
round(100.0 * SUM(CASE WHEN availability_365 BETWEEN 1 AND 364 THEN 1 ELSE
0 END) / count(*), 2) as Partially_Available_Percent,
count(*) as Total_Listings
FROM listings;
```

OutPut:

	Fully_Available_Percent	Fully_Booked_Percent	Partially_Available_Percent	Total_Listings
1	3.61	4.18	92.21	8660

Analysis: The vast majority of listings (92.21%) are partially available, meaning they are booked for some period, but not others, which reflects typical short-term rental behaviour. Only 3.61% of listings are fully available year-round, suggesting most hosts are actively managing bookings rather than leaving properties idle. Interestingly, 4.18% of listings are fully booked with 0 availability, indicating a small but highly demanded segment. Overall, the high partial availability rate suggests a healthy and active rental market across Chicago's 8,660 total listings.